Fast MPC & MPC-BAM Overview

Updated 2 months ago

> **Early Access feature**
>
> This feature is currently in Early Access. If you are interested in joining the Early Access phase, contact your Fireblocks Customer Success Manager.

Why use MPC-BAM?

As part of the Fast MPC product family, MPC-BAM delivers signing speeds up to 10 times faster than MPC-CMP while maintaining Fireblocks' security guarantees, enabling you to execute latency-sensitive operations without compromise. This includes, but is not limited to, the following use cases:

- **High-frequency trading:** Execute trades faster with sub-second signing times
- **Automated market making:** Respond to market conditions with minimal latency
- **Payment processing:** Process high volumes of transactions quickly
- **Programmatic vault operations:** Execute automated workflows with reduced delays
- **DeFi protocol interactions:** Interact with smart contracts more efficiently

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How MPC-BAM works

MPC-BAM is a novel 2/2 ECDSA protocol that achieves two rounds (a single back-and-forth communication). This reduction in cryptographic rounds enables sub-second signing times for ECDSA SECP256K1 transactions.

Built on proven research

Fireblocks built MPC-BAM based on peer-reviewed academic research published by Fireblocks researchers and academic partners (Two-Round 2PC ECDSA at the Cost of 1 OLE, by Michael Adjedj, Constantin Blokh, Geoffroy Couteau, Arik Galansky, Antoine Joux, and Nikolaos Makriyannis (2024)). Trail of Bits independently audited the protocol for security assurance.

API user. When you enable MPC-BAM for a specific API user paired with a co-signer:

1. The co-signer generates new 2/2 keys in the background. Once provisioning completes, the API user signs all transactions with MPC-BAM.
2. If any issues occur with MPC-BAM, the co-signer automatically uses your existing MPC-CMP keys for signing.
3. If you disable MPC-BAM, the API user reverts to MPC-CMP signing. Your MPC-BAM keys remain but are not used.

You can enable MPC-BAM for high-throughput API users while keeping others on MPC-CMP, allowing you to test and adopt gradually.

FAQ

What happens to my existing MPC-CMP keys if I enable MPC-BAM?

Your existing MPC-CMP keys remain active and continue to function as an automatic fallback. If any issues occur with MPC-BAM, the system automatically uses your MPC-CMP keys for signing. You can disable MPC-BAM at any time for any API user—when disabled, the API user reverts to MPC-CMP for all signing operations.

What happens if MPC-BAM has issues?

The system includes automatic failover to MPC-CMP. If MPC-BAM encounters any problems (eg key provisioning fails, service unavailable), your API user automatically falls back to MPC-CMP signing with no manual intervention required.

Does MPC-BAM work with mobile co-signers?

No. MPC-BAM supports only API co-signers (SGX, AWS Nitro, GCP). Mobile co-signers continue to use MPC-CMP for all signing operations.

What algorithms does MPC-BAM support?

MPC-BAM supports only ECDSA SECP256K1, which Bitcoin, Ethereum, and most EVM-compatible chains use. EdDSA algorithms (used by Solana, Cardano, and other chains) are not supported by MPC-BAM and continue to use MPC-CMP.

How long does key provisioning take?

No. MPC-BAM does not change transaction fees or blockchain costs. The performance improvement is in the signing process itself, not in on-chain transaction execution.

Can I enable MPC-BAM for some API users and not others?

Yes. MPC-BAM is configured per API user. You can enable it for high-throughput API users while keeping others on MPC-CMP, allowing you to test and adopt gradually.

Does background key generation require the Owner?

No. Your co-signer and the cloud co-signer re-shuffle the MPC-CMP key shares, similar to how a new signer joining a workspace gets keys from the Owner. However, since MPC-BAM keys are provisioned from the co-signer to itself, this happens securely in the background without human involvement.

How do new co-signers get MPC-BAM keys?

The initial setup process begins with the generation of MPC-CMP key shares as usual. After that is complete, you enable MPC-BAM for the co-signer.

What about backups?

MPC-BAM doesn't change your underlying private keys, only the signing protocol. Your existing DRS backup packages remain valid because MPC-BAM keys reconstruct to the same private key and public key as your MPC-CMP keys. However, back up your co-signer's database to simplify recovery.

Do I need my passphrase to enable MPC-BAM?

No. Since mobile co-signers do not currently support MPC-BAM, your passphrase is not required to enable it.

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